

In [1]: `import pandas as pd` *#Step 1: Import Libraries*

In [5]: `df=pd.read_csv('C:/Users/hp/Desktop/hr_analytics_sample_data.csv')` *#Step 2:*
`df.head()`

Out[5]:

	Employee_ID	Name	Age	Department	Gender	Joining_Date	Salary	Experience_Ye
--	-------------	------	-----	------------	--------	--------------	--------	---------------

0	E001	Ravi Kumar	29	IT	Male	2018-06-12	65000	
---	------	------------	----	----	------	------------	-------	--

1	E002	Meena Roy	34	HR	Female	2015-03-22	58000	
---	------	-----------	----	----	--------	------------	-------	--

2	E003	Arjun Das	41	Finance	Male	2010-11-01	72000	
---	------	-----------	----	---------	------	------------	-------	--

3	E004	Sita Verma	25	Marketing	Female	2021-01-05	45000	
---	------	------------	----	-----------	--------	------------	-------	--

4	E005	Ali Khan	38	Operations	Male	2014-08-17	68000	
---	------	----------	----	------------	------	------------	-------	--



In [6]: `df.head()` *# View first 5 rows* *Step 3: Basic Data Exploration*

Out[6]:

	Employee_ID	Name	Age	Department	Gender	Joining_Date	Salary	Experience_Ye
--	-------------	------	-----	------------	--------	--------------	--------	---------------

0	E001	Ravi Kumar	29	IT	Male	2018-06-12	65000	
---	------	------------	----	----	------	------------	-------	--

1	E002	Meena Roy	34	HR	Female	2015-03-22	58000	
---	------	-----------	----	----	--------	------------	-------	--

2	E003	Arjun Das	41	Finance	Male	2010-11-01	72000	
---	------	-----------	----	---------	------	------------	-------	--

3	E004	Sita Verma	25	Marketing	Female	2021-01-05	45000	
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4	E005	Ali Khan	38	Operations	Male	2014-08-17	68000	
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In [7]: `df.shape` *# Check (rows, columns)*

Out[7]: (7, 10)

In [8]: `df.columns` *# Column names*

Out[8]: Index(['Employee_ID', 'Name', 'Age', 'Department', 'Gender', 'Joining_Date', 'Salary', 'Experience_Years', 'Performance_Rating', 'Status'], dtype='object')

In [9]: `df.info()` *# Data types + missing values*

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7 entries, 0 to 6
Data columns (total 10 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Employee_ID           7 non-null      object
1   Name                   7 non-null      object
2   Age                    7 non-null      int64
3   Department             7 non-null      object
4   Gender                 7 non-null      object
5   Joining_Date           7 non-null      object
6   Salary                 7 non-null      int64
7   Experience_Years       7 non-null      int64
8   Performance_Rating     7 non-null      int64
9   Status                 7 non-null      object
dtypes: int64(4), object(6)
memory usage: 692.0+ bytes
```

```
In [10]: df.describe()           # Summary stats for numerical columns
```

```
Out[10]:
```

	Age	Salary	Experience_Years	Performance_Rating
count	7.000000	7.000000	7.000000	7.000000
mean	32.000000	59428.571429	6.000000	3.714286
std	5.887841	10244.626931	4.242641	1.112697
min	25.000000	45000.000000	1.000000	2.000000
25%	28.000000	52500.000000	3.000000	3.000000
50%	30.000000	61000.000000	5.000000	4.000000
75%	36.000000	66500.000000	8.500000	4.500000
max	41.000000	72000.000000	13.000000	5.000000

```
In [11]: df.isnull().sum()           # Check for missing values           Step 4: Data Cleaning
```

```
Out[11]: Employee_ID           0
          Name                 0
          Age                  0
          Department           0
          Gender               0
          Joining_Date         0
          Salary               0
          Experience_Years     0
          Performance_Rating   0
          Status               0
          dtype: int64
```

```
In [12]: df.duplicated().sum()       # Check for duplicates
```

```
Out[12]: np.int64(0)
```

```
In [13]: df.columns = df.columns.str.replace(" ", "_")           # Clean column names (optional)
```

```
In [16]: df['Joining_Date'] = pd.to_datetime(df['Joining_Date'])           # Convert Joining_Date
```

```
In [19]: from datetime import datetime          # Current Year Experience      Step
current_year = datetime.now().year
df['Calculated_Experience'] = current_year - df['Joining_Date'].dt.year
```

```
In [20]: df.groupby('Department')['Salary'].mean()    # Average salary by department    S
```

```
Out[20]: Department
Finance      59500.0
HR           58000.0
IT           63000.0
Marketing    45000.0
Operations   68000.0
Name: Salary, dtype: float64
```

```
In [21]: df['Department'].value_counts()             # Count of employees by department
```

```
Out[21]: Department
IT          2
Finance     2
HR          1
Marketing   1
Operations  1
Name: count, dtype: int64
```

```
In [22]: df['Gender'].value_counts()                 # Gender distribution
```

```
Out[22]: Gender
Male       4
Female     3
Name: count, dtype: int64
```

```
In [23]: df.groupby('Department')['Performance_Rating'].mean()    # Average performance r
```

```
Out[23]: Department
Finance      2.5
HR           5.0
IT           3.5
Marketing    4.0
Operations   5.0
Name: Performance_Rating, dtype: float64
```

```
In [24]: high_perf = df[df['Performance_Rating'] >= 4]    # Employees with performance ra
```

```
In [25]: it_dept = df[df['Department'] == 'IT']           # Employees in IT department
```

```
In [26]: resigned = df[df['Status'] == 'Resigned']        # Resigned employees
```

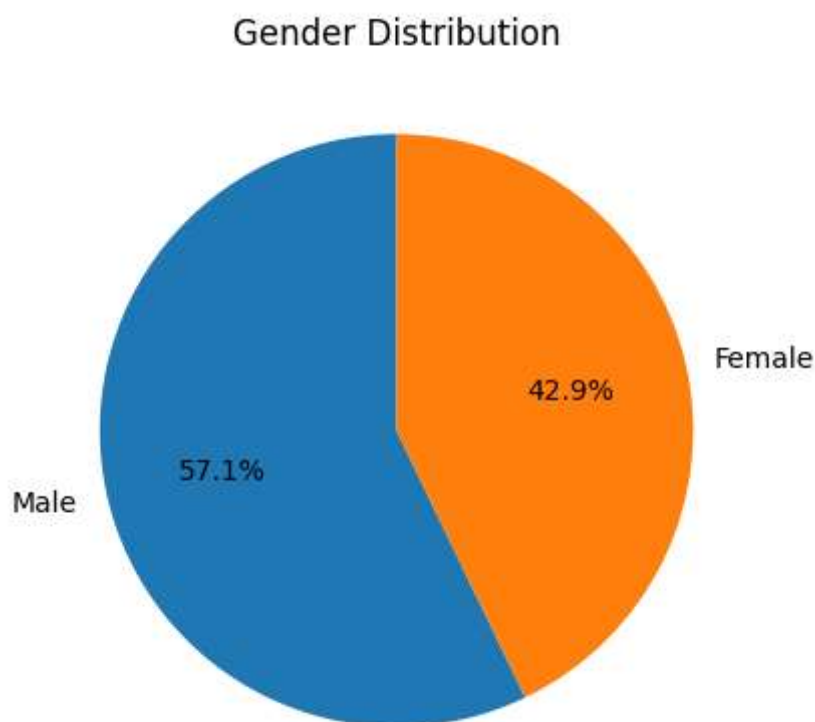
```
In [27]: df.corr(numeric_only=True)                     #Step 8: Correlation (Optional)
```

Out[27]:

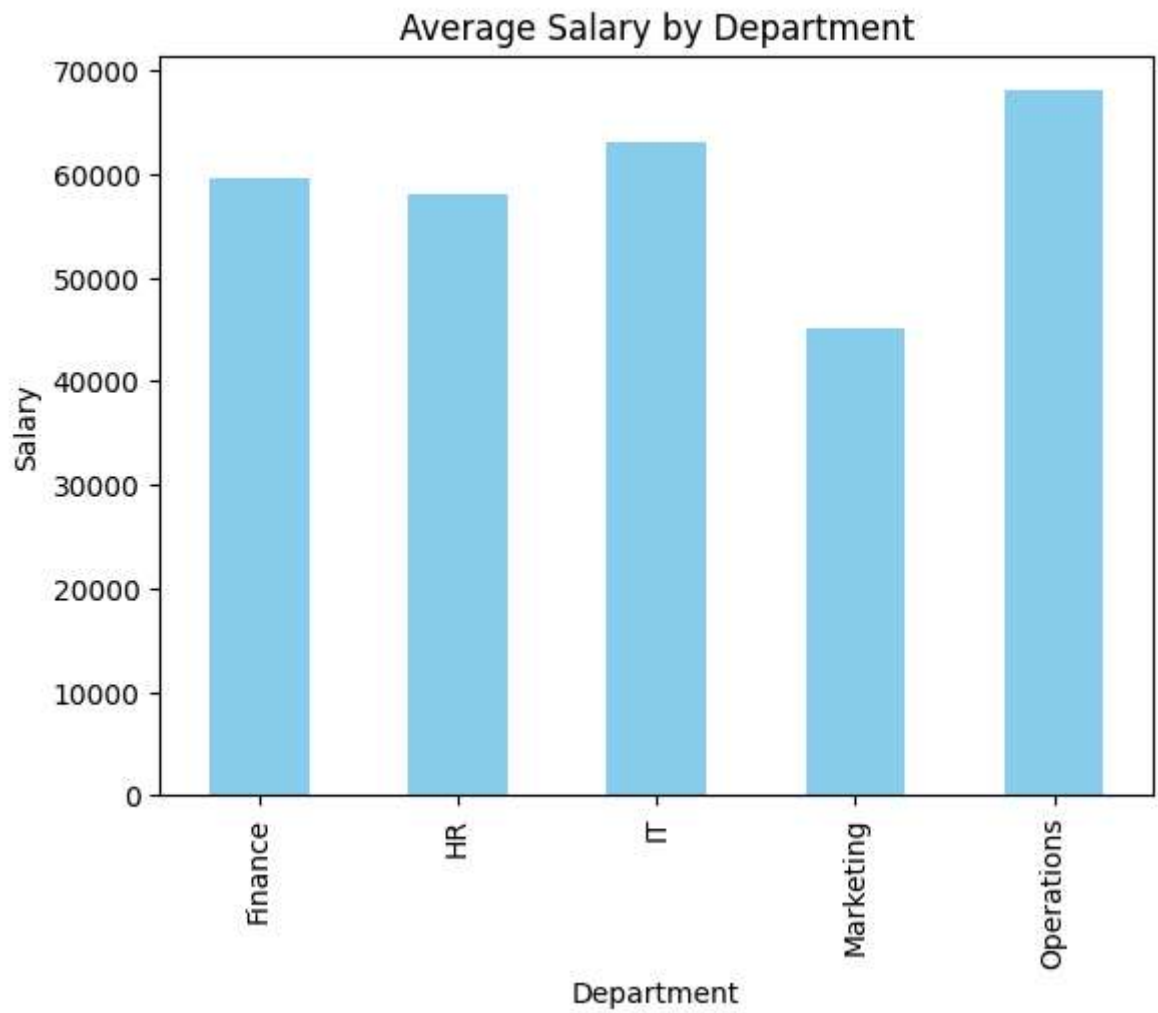
	Age	Salary	Experience_Years	Performance_Rating	Calculated_Experience
Age	1.000000	0.842745	0.967441	0.279839	
Salary	0.842745	1.000000	0.851274	0.261088	
Experience_Years	0.967441	0.851274	1.000000	0.353049	
Performance_Rating	0.279839	0.261088	0.353049	1.000000	
Calculated_Experience	0.967441	0.851274	1.000000	0.353049	

In [28]: `import matplotlib.pyplot as plt` *#Step 9: Simple Visualizations*
`import seaborn as sns`

In [29]: *# Gender distribution pie chart*
`df['Gender'].value_counts().plot.pie(autopct='%1.1f%%', startangle=90)`
`plt.title("Gender Distribution")`
`plt.ylabel("")`
`plt.show()`



In [30]: *# Bar chart of average salary per department*
`df.groupby('Department')['Salary'].mean().plot(kind='bar', color='skyblue')`
`plt.title("Average Salary by Department")`
`plt.ylabel("Salary")`
`plt.show()`



```
In [36]: df.to_csv(r'C:\Users\hp\Desktop\cleaned_hr_analytics_sample_data.csv', index=False)
```

```
In [ ]:
```